
Vaccination Slot Booking & Tracking Platform

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ABSTRACT

Vaccination Slot Booking and Tracking Platform

- The **Vaccination Slot Booking and Tracking Platform** is a web-based system for easy vaccination appointment management.
- Users can **view available vaccination slots and book** a suitable date and time.
- Users can **track their booking status and vaccination history** through the platform.
- Admins can **manage vaccination centres, vaccines, slots, and appointments**.
- The system stores booking and vaccination details in a **centralised database**, reducing manual work and scheduling errors.
- The platform provides a **simple, secure, and user-friendly** way to manage vaccination appointments and can be enhanced with reminders and notifications in the future.

Problem Statement

Finding available vaccination slots can be difficult and time-consuming.

Manual booking processes may lead to delays, confusion and missed appointments.

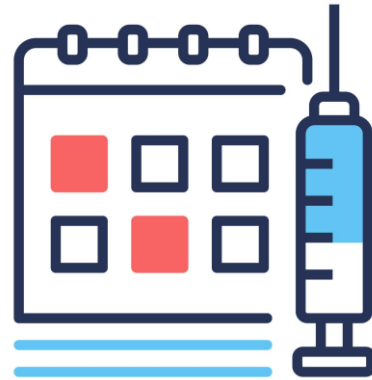
Users need a simple way to discover vaccination centers and available time slots.

Appointment status, vaccination history and reminders should be easy to track.

Existing System

Existing vaccination systems mainly focus on appointment booking, while real-time slot availability, vaccination history, dose tracking, and centralised user management are limited or handled separately.

1. Co-win –India
2. NHS booking system -UK



MY Proposed System

Vaccination Slot Booking and Tracking Platform integrates slot booking, real-time availability, vaccination history, dose reminders, and vaccination status tracking into a single platform.

Objectives

Provide convenient and organized online vaccination slot booking.

Display vaccination center and slot availability clearly.

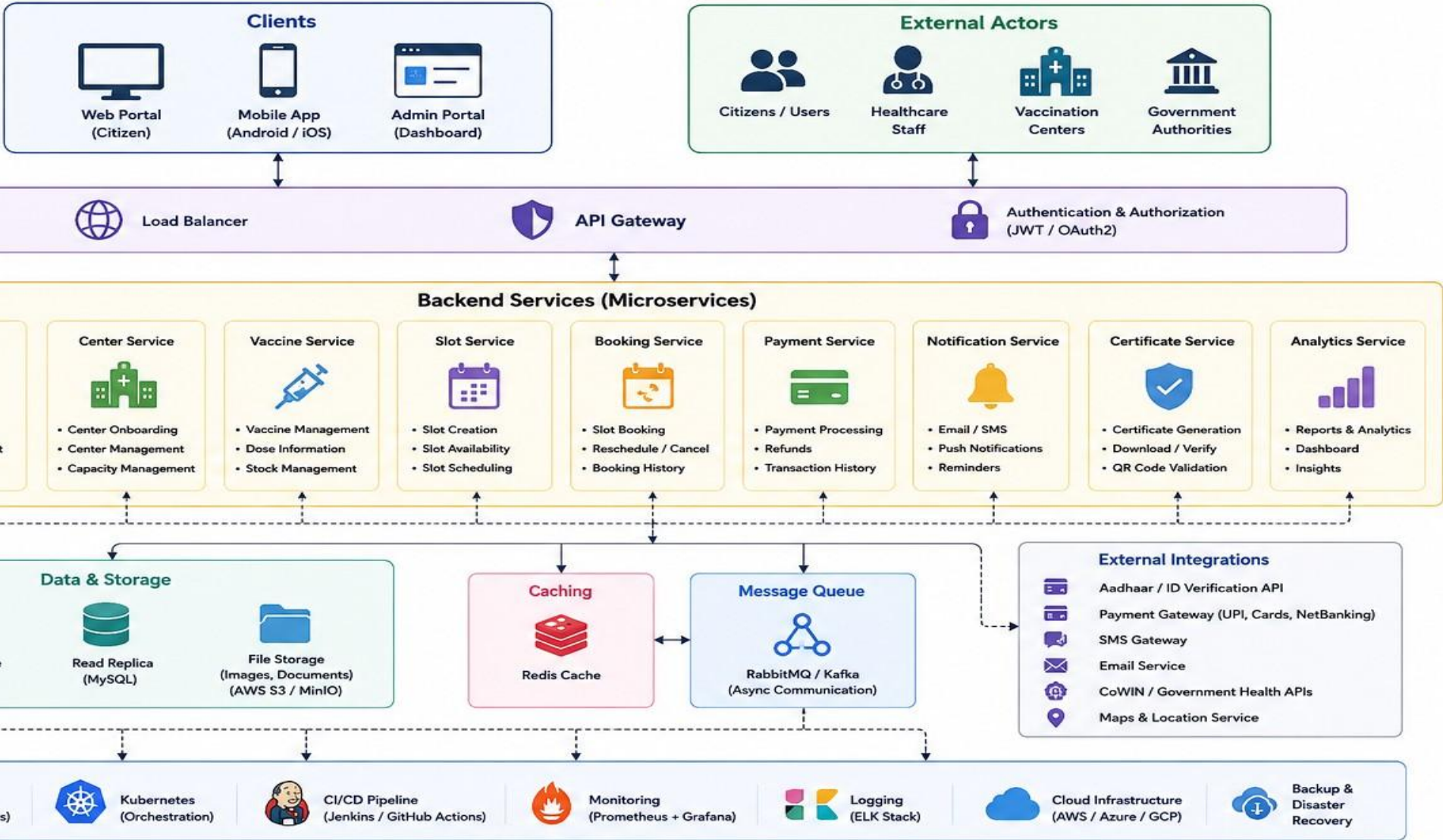
Reduce waiting time and manual coordination.

Enable appointment and vaccination-status tracking.

Support timely notifications and efficient administration.

Vaccination Slot Booking & Tracking Platform

System Architecture

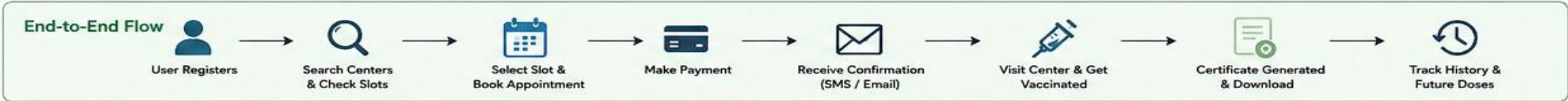


Key Features

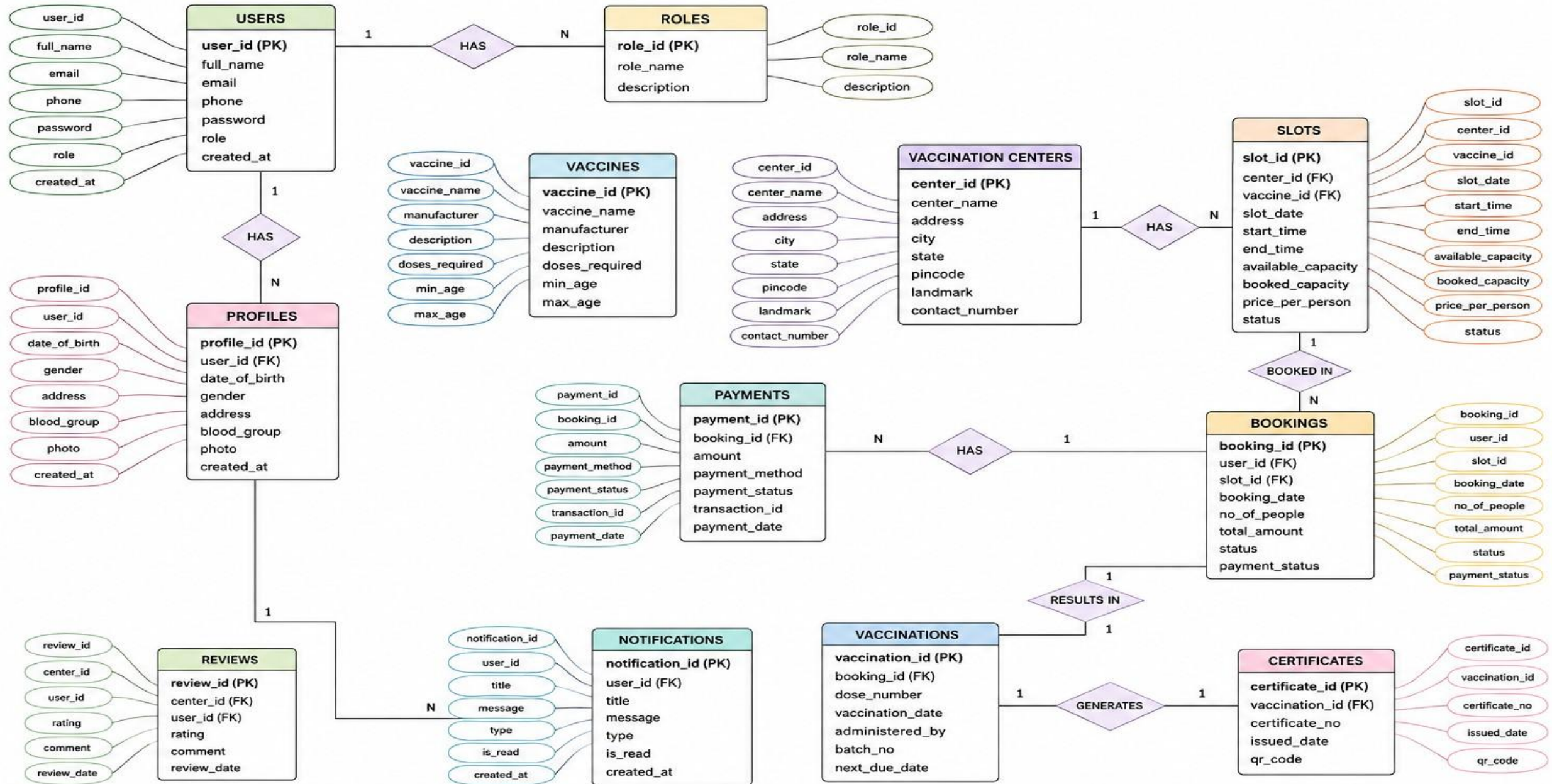
- ✓ Real-time Slot Availability
- ✓ Easy Slot Booking
- ✓ Multiple Payment Options
- ✓ Booking & Appointment Tracking
- ✓ Reschedule / Cancel
- ✓ Notifications & Reminders
- ✓ Digital Vaccination Certificate
- ✓ Admin & Center Dashboard
- ✓ Reports & Analytics
- ✓ Secure & Scalable System

Technology Stack

- Frontend**
 - React.js / Next.js
- Mobile**
 - Flutter / React Native
- Backend**
 - Spring Boot (Java)
- Database**
 - MySQL
- Cache**
 - Redis
- Message Queue**
 - RabbitMQ / Kafka
- Cloud**
 - AWS / Azure / GCP



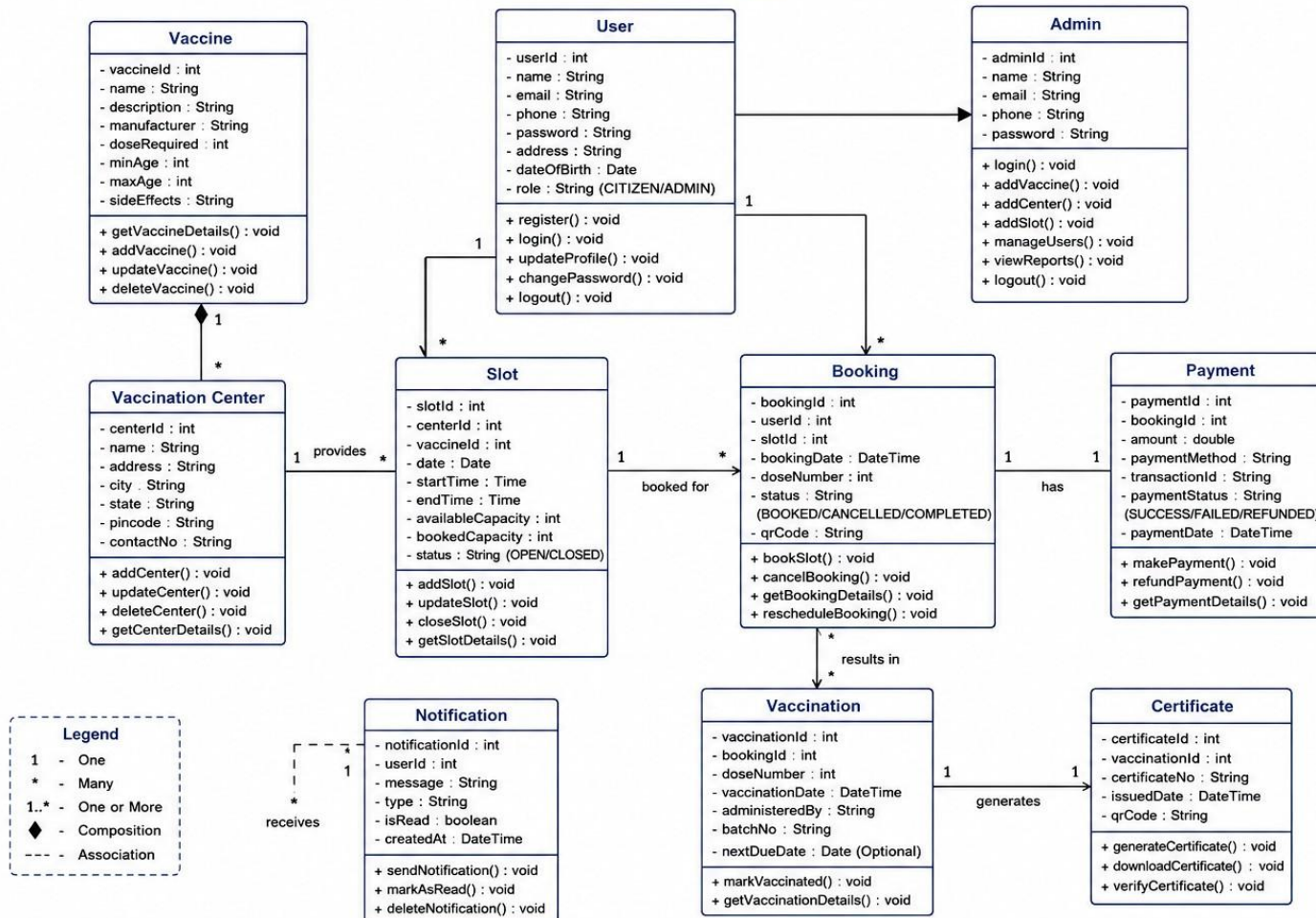
ER DIAGRAM – VACCINATION SLOT BOOKING & TRACKING PLATFORM



Class Diagram

Vaccination Slot Booking & Tracking Platform

Class Diagram



Requirements / SRS

User: Registration, Login, Profile Management and Center Search.

Booking: View Slots, Book/Cancel Appointment and Track Status.

Notification: Appointment reminders and booking-status updates.

Admin: Manage Users, Centers, Vaccines, Slots and Records.

Security: Authentication, authorization and protected user data.

Technology Stack

Frontend: React.js / HTML, CSS, JavaScript

Backend: Java / Spring Boot

Database: MySQL

API Testing: Postman

Development & Version Control: VS Code, Git & GitHub

Conclusion & Future Enhancement

Provides a simple and organized vaccination booking experience.

Improves appointment visibility and tracking for users.

Reduces administrative effort and manual coordination.

Future: Mobile app, SMS/Email notifications, QR verification and analytics dashboard.